IEEE DATAPORT DOI: 10.21227/Z7HV-E014 • FINAL PROJECT DEFENSE

15-Minute Defense Plan

SANADA: Reproducible Synthetic Acoustic Dataset for Intractable Mixtures

Benchmarking Variable-K Source Separation, Density Estimation, and Tagging up to K=32 with Full-Stack Web Architecture.

Frontend & UI Lead
Nikhilesh

Task 1: Counting
Shravan

Task 2: Tagging
Manaswini

Task 3: Separation
Srikari

Data & Backend Lead
Eshan

Real-World Scenes vs. 2-Speaker Lab Capping

Standard benchmarks cap evaluations at low speaker counts without clean ground-truth reference stems.

Benchmark Dataset	Acoustic Domain	Cardinality (K)	Isolated Stems?	Variable-K Design?
WSJ0-2mix / 3mix	Read Speech	K = 2, 3	**Yes**	**No**
LibriMix / WHAMR!	Noisy Speech	K = 2, 3	**Yes**	**No**
AudioSet / FSD50K	General Audio	Uncontrolled	**No (Mixed Only)**	**No**
FUSS Benchmark	Environmental	K ≤ 4	**Partial**	**Yes**
SANADA Suite (Ours)	**Speech & 32 Environmental**	**K = 2 to 32**	**Yes (27,112 Stems)**	**Yes (Fixed + Var)**

Physical intractability solved: synthetic mixing delivers clean master stems for SI-SDR evaluation up to K=32.

ESHAN• GENERATIVE MIXING ENGINE

The 3-Phase Generative Pipeline

Acoustic curation, strict pre-mix isolation, and anti-clipping algorithmic superposition.

Phase 1 Domain Acquisition

Two isolated acoustic wings:

- **Wing 1:** Studio organizational lectures.
- **Wing 2:** 32 verified FSD classes.
- **Standard:** 16 kHz 16-bit mono WAV.

Phase 2 Pre-Mix Partitioning

Strict zero-leakage protocol:

- **Seed S=42:** 70/15/15 disjoint stem split.
- **Buffer:** 15.0s edge cut removes noise.
- **Quotas:** 1000/500/500 stems per class.

Phase 3 Algorithmic Mixing

Full polyphonic superposition:

- **Overlap:** 100% active 8.0s window.
- **Attenuation:** -3.0 dBFS scaling.
- **Manifest:** metadata.csv tracking.

Anti-Clipping Theorem (Eq. 1): `x_mix(t) = s_1(t) + Σ 10^(-3.0/20) • s_i(t)` Eliminates integer overflow without DRC artifacts

ESHAN• DATASET CURATION & INTEGRITY AUDIT

Dataset Footprint and Signal Physics

Standardized 8.0–second single-channel waveforms spanning speech and environmental corpora.

27,112
Master Pre-Mix Stems (60.25h)

15,600
Synthesized Mixtures (34.66h)

128,000
Samples per Clip (16 kHz Mono)

18.4 dB
Mean WADA-SNR (RT60 ≈ 0.47s)

Corpus Subsets & K Densities

- **Org-Mix-4:** $K \in \{2, 3, 4, \text{Variable}\}$ organizational speech.
- **Org-Mix-9:** $K \in \{6, 9, \text{Variable}\}$ high-density speech.
- **FSD-Mix:** $K \in \{2, 3, 15, 20, 25, 32\}$ environmental events.

Physical Spectral Masking

- **Spectral Centroid:** Drops from 2112.8 Hz to 1503.5 Hz at K=9.
- **RMS Accumulation:** Escalates from 0.066 to 0.204.
- **Masking Impact:** Complete loss of temporal silence gaps.

ESHAN • SPLIT AUDIT & TENSOR PIPELINES

Pre-Mix Zero-Leakage & DataLoaders

Partitioning stems before mixing guarantees zero speaker or class leakage across splits.

zero_leakage_audit.py — TEST PASSED

```
# Algorithm 3: Deterministic Pre-Mix File Lock random.seed(42)
assert set(train_stems).isdisjoint(val_stems) assert
set(train_stems).isdisjoint(test_stems) # PyTorch Multi-Threaded
Streaming loader = DataLoader( dataset, batch_size=32,
num_workers=4, pin_memory=True ) # Throughput: ≥ 10 batches/second
```

Deterministic Seed Protocol

PRNG initialized with S=42 for exact reproduction.

Zero Speaker Leakage

Test mixtures contain exclusively unseen speakers.

Batching Pipeline Throughput

Over 10 batches per second feeding PyTorch models.

SHRAVAN • TASK 1: CONFORMER REGRESSION

Acoustic Density Estimation (Counting)

Estimating the active source count K from single-channel 8.0s mixtures via continuous regression.

74.8%

Exact Accuracy on Org-Mix-4

100.0% Relaxed Accuracy (±1 source)

Conformer Dual-Context Backbone

Interleaved multi-head self-attention with depthwise convolutions.

Continuous MSE Optimization

Linear projection unit trained with Mean Squared Error loss.

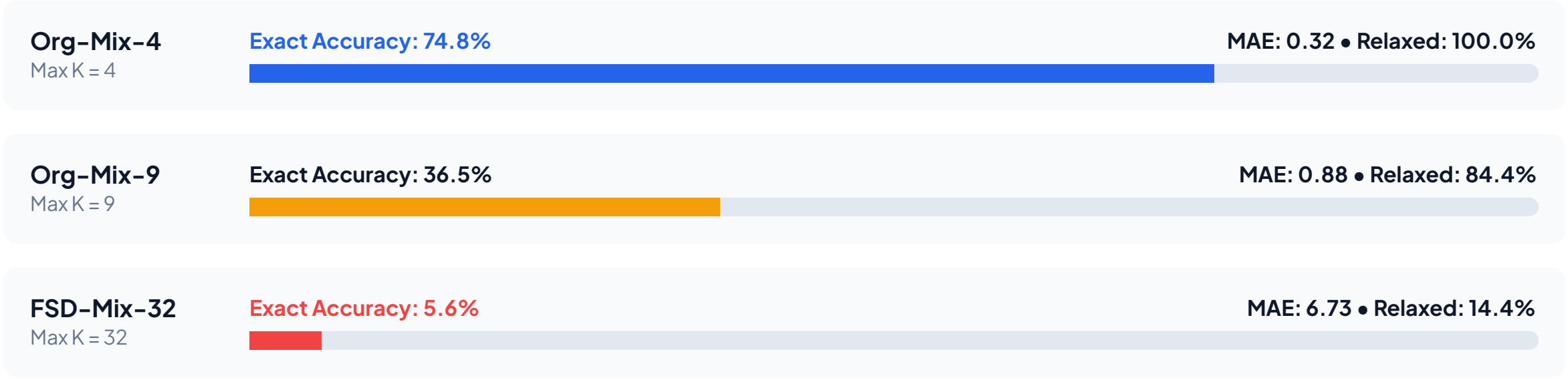
Inference Clamping & Rounding

```
K_hat = round(clamp(K_pred, min=1, max=K_max))
```

SHRAVAN • EMPIRICAL COUNTING RESULTS

Source Counting Degradation Trajectory

Higher cardinality and environmental diversity eliminate spectral sparsity cues.



Architectural Bottleneck: At K > 9, total spectral collision eliminates harmonic sparsity cues needed for scalar regression.

MANASWINI • TASK 2: MULTI-LABEL TAGGING

Multi-Label Tagging & Long-Tail Skew

Conformer classification powered by Asymmetric Loss to counter extreme negative-label imbalance.

32-Class Long-Tail Imbalance

FSD_Master stem count varies heavily by acoustic duration:

- **Sustained (Saxophone, Flute):** ~100–110 stems.
- **Medium (Acoustic Guitar, Applause):** ~40–48 stems.
- **Transient (Shatter, Squeak, Cough):** ~11–12 stems.

Algorithm 5 draws classes uniformly to test density over frequency.

Asymmetric Loss (ASL) Solution

Resolves extreme positive/negative label skew:

- **30 Negative Classes:** Only 2 active classes at K=2.
- **Margin Shift:** $p_m = \max(p - m, 0)$ zeroes easy negatives.
- **Decoupled Gammas:** $\gamma_{neg} > \gamma_{pos}$ prevents saturation.

Prevents zero-gradient collapse on rare acoustic classes.

MANASWINI • TAGGING PERFORMANCE EVALUATION

Acoustic Tagging Benchmark Results

Macro-averaged precision, recall, and ROC-AUC metrics across speech and environmental scene complexity.

4 Classes

Org-Mix-4

mAP (Macro): 0.9902

F1 Score: 0.9449

ROC-AUC: 0.9765

Near-perfect speaker identification.

9 Classes

Org-Mix-9

mAP (Macro): 0.9122

F1 Score: 0.8305

ROC-AUC: 0.8744

Robust separation across talkers.

32 Classes

FSD-Mix

mAP (Macro): 0.6265

F1 Score: 0.6849

ROC-AUC: 0.6161

Transient sounds masked by sustained audio.

SRIKARI • TASK 3: CONV-TASNET BASELINE

Universal Monaural Source Separation

Reconstructing isolated waveforms from single-channel mixtures using time-domain convolutional masking.

Stage 01

1D-Conv Encoder

Phase-preserving waveform basis representation.

Stage 02

Dilated TCN Masks

Estimates K source masks via negative SI-SDR loss.

Stage 03

1D-Deconv Decoder

Synthesizes reconstructed time-domain audio stems.

Fixed-K Separation Fidelity Curve (SI-SDR dB)

+10.59 dB	+5.08 dB	+4.24 dB	-3.40 dB	-8.57 dB
K = 2 (Org)	K = 3 (Org)	K = 4 (Org)	K = 6 (Collapse)	K = 9 (Intractable)

Theoretical Ceilings Exposed by SANADA

Standard deep learning separation paradigms suffer from three fundamental architectural limits.

Bottleneck 1

The Complexity Ceiling

SI-SDR falls below 0 dB when K > 4 speakers. Time-domain masking collapses under heavy spectral saturation.

Bottleneck 2

O(K!) Permutation Wall

uPIT checks K! permutations. At K=15, evaluating 1.3T combinations triggers CUDA Out-Of-Memory crashes on A100s.

Bottleneck 3

Zero-Padding Ambiguity

Fixing output nodes to N=9 and zeroing inactive channels drops variable K≤4 SI-SDR from 4.24 dB to 1.98 dB.

Research Mandate: High-density listening demands target-query models and dynamic-cardinality routing.

NIKHILESH & ESHAN • INTERACTIVE WEB ARCHITECTURE

Multi-Tier Cloud System Architecture

Production full-stack listening portal connecting Vite/React client to Express API and PyTorch workers.

Tier 1: Frontend (Nikhilesh)

React + Web Audio Portal

- **Mic Engine:** MediaRecorder 8.0s audio.
- **Visualizer:** Canvas Mel-spectrograms.
- **Stem Mixer:** Solo, Mute, Volume faders.

Tier 2: Backend (Eshan)

Express REST Gateway

- **Route:** POST /api/predict multipart.
- **Validation:** 16 kHz 16-bit mono checks.
- **Controls:** /api/train and /api/stats.

Tier 3: Inference Engine

PyTorch & ONNX Runtime

- **Model 1:** Conformer Count Regressor.
- **Model 2:** Conformer + ASL Tagger.
- **Model 3:** Conv-TasNet Stem Separator.

```
Live Demonstration: Org_Mix_4_Test_042.wav • K=3 • Stems Reconstructed   200 OK (412ms)
```